

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Regular and Supplementary Examination December 2022 (2019 Scheme)

Course Code: AIT 307**Course Name: INTRODUCTION TO ARTIFICIAL INTELLIGENCE**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- | | | |
|----|--|---|
| 1 | How the definitions of artificial intelligence organized into four categories? | 3 |
| 2 | What is a Turing test approach? | 3 |
| 3 | What are the disadvantages of BFS search algorithm? | 3 |
| 4 | How the vacuum cleaner world formulated as a toy problem? | 3 |
| 5 | Explain the six elements of the search problem for the game tic-tac-toe. | 3 |
| 6 | What are the components of a map colouring CSP problem? | 3 |
| 7 | Write the PEAS description of the Wumpus world. | 3 |
| 8 | What are the steps involved in Knowledge Engineering in FOL? | 3 |
| 9 | What are the components of the agents to be learned? Explain with an example. | 3 |
| 10 | Describe the various forms of learning? | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

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|----|---|---|
| 11 | a) Discuss about any five disciplines that contributed ideas, viewpoints, and techniques to AI | 7 |
| | b) Explain the properties of task environments. | 7 |
| 12 | a) Discuss about any five applications of AI. | 7 |
| | b) Explain the structure of Simple reflex agents and Model-based reflex agents with the help of diagrams. | 7 |

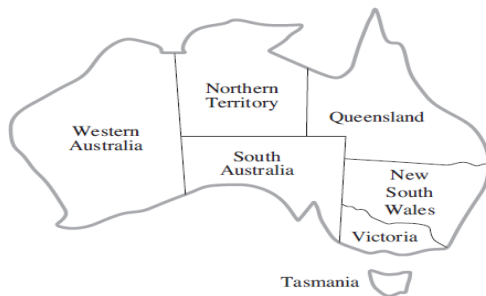
Module -2

- | | | |
|----|---|---|
| 13 | a) What are the components of a well-defined AI problem? Explain each component based on 8-Queen problem. | 7 |
| | b) Describe any three uninformed search strategies. | 7 |

- 14 a) What are the real-world problems? Discuss about any two real world problems. 7
 b) Explain the recursive best first search algorithm with diagram. 7

Module -3

- 15 a) Explain the MINIMAX algorithm with an example. 7
 b) What is a constraint satisfaction problem? Formulate the job-shop scheduling as a CSP problem. 7
- 16 a) Solve graph colourings problem for the following graph given below using backtracking search for CSP. 7



- b) Explain the following terms. 7
 a. Constraint Graph
 b. Topological Sort

Module -4

- 17 a) What are the basic elements of the syntax of a First Order Logic? 7
 b) Convert the following sentences into FOL. 7
- Brothers are siblings.
 - One's mother is one's female parent.
 - Onion and Potato are vegetables.
- 18 a) What are the steps involved in the Knowledge representation of a FOL? Explain with an example. 7
 b) Explain the concept of unification with example. 7

Module -5

- 19 a) How is best hypothesis selected from alternatives 7
 b) Explain Decision Tree Learning Algorithm with a suitable example. 7
- 20 a) Explain the concept of regression with linear models 7
 b) Explain in detail about reinforcement learning. 7
